

ARAVIND V

8667097747 | aravind.offwork@gmail.com | linkedin.com/in/aravind-v-30415b302 | github.com/ARAVIND729
aravind007.netlify.app

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate from Agni College of Technology with hands-on experience in IoT and embedded system development. Proficient in ESP32 and Arduino-based systems, sensor interfacing, MQTT communication, and cloud platform integration through internship training and project work. Looking to contribute practical embedded systems and automation skills to an entry-level role and grow within a results-driven engineering environment.

EDUCATION

Bachelor of Engineering – Electronics and Communication Engineering <i>Agni College of Technology, Anna University, Chennai CGPA: 8.2 / 10.0</i>	2022 – 2026
Higher Secondary Certificate (HSC) <i>Valluvar Gurukulam Higher Secondary School 70.8%</i>	2021 – 2022

INTERNSHIP EXPERIENCE

Emertxe Information Technologies – Embedded Systems Intern	January 2025 – February 2025
- Implemented MQTT-based communication for real-time sensor data acquisition in IoT systems, ensuring stable and low-latency data transfer. - Designed and optimized data pipelines to improve system reliability and transmission efficiency. - Enabled continuous monitoring for industrial IoT applications through robust communication protocols.	

PROJECTS

IoT-Enabled Smart EV Vehicle Monitoring System with Real-Time Video Streaming

- Built an ESP32-based vehicle monitoring system integrating 4+ sensors (MPU6050, MQ2, DHT22, VL53L0X) for real-time environmental and motion data acquisition.
- Implemented real-time data transmission to ThingSpeak cloud with update latency under 30 seconds; applied ML models (Logistic Regression, SVM, Random Forest) in MATLAB and Python for predictive analytics.
- Integrated ESP32-CAM for live video streaming at 15-20 FPS and GSM/RF modules for remote alert notifications with response time under 5 seconds.

PCB Design – Headphone Amplifier

- Designed a low-noise op-amp headphone amplifier in KiCad with gain of 5 (14 dB) using a +/-12V supply, covering full audio bandwidth of 20 Hz to 20 kHz.
- Delivered stable output of approximately 100 mW into 32 Ohm headphones with low distortion through optimized PCB layout.

Smart Energy Management System Using ESP32

- Developed an ESP32-based IoT system for automated appliance control using Blynk, reducing manual intervention by 60% through rule-based automation.
- Built a real-time monitoring dashboard for remote access, scheduling, and energy tracking.

IoT Home Automation System (Simulation)

- Designed a home automation system using Arduino, PICSimLab, and Blynk integrating 4+ modules for lighting, temperature sensing, water-level monitoring, and climate control.
- Achieved response time under 2 seconds and greater than 95% system reliability during simulation and continuous testing.

TECHNICAL SKILLS

Embedded Systems: ESP32, Arduino, Sensor Interfacing, Embedded C

IoT Platforms: Blynk IoT, ThingSpeak, MQTT, Real-Time Data Monitoring

Communication Protocols: UART, SPI, I2C

Programming: C, MATLAB, Python

Design & Simulation Tools: KiCad, LTspice, PICSimLab

Automation (RPA): UiPath, Workflow and Rule-Based Automation

CERTIFICATIONS

- Embedded Systems and MATLAB & Simulink – NIELIT
- NPTEL Elite + Silver – Soft Skills and Personality Development

ACTIVITIES & LEADERSHIP

- Department Event Coordinator – Agni College of Technology
- Member – Rotaract Club, Agni College of Technology
- Volunteer – Intercollegiate Cultural Events